

AADITYA SAPKOTA

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SUMMARY

Ambitious AI Researcher and Data Science undergraduate passionate about applying machine learning with advanced mathematics to real-world challenges. Skilled in programming, data analysis, and visualization, with experience in event management and community contributions. Dedicated to bridging theory and practical impact through innovative AI research and open-source collaboration.

SKILLS

- **Software & Tools:** Python, R, SQL, JavaScript, HTML/CSS, Git/Github, Docker,scikit-learn
 - **Data & AI:** Machine Learning, Deep Learning, Data Analysis, Data Visualization, NLP
 - **Soft Skills:** Leadership ,Teamwork, Problem-Solving, Critical Thinking, Communication
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PROJECTS

1. Nepali News Analytics

- Built a full Nepali news pipeline: scraping, cleaning, classification, sentiment, and summarization in one consistent workflow.
- Compared classical models and transformer approaches; published side-by-side HTML reports for faster model decisions.
- Automated daily data refresh so experiments stay current without manual prep.

2. NextGen Code Generator

- Developed a TypeScript CLI that scaffolds API endpoints, types, and tests, cutting repetitive setup work.
- Designed a modular template + AST system so new generation patterns plug in quickly.
- Added lint and snapshot tests to ensure generated code is clean on first commit.

3. Nepali Poetic LLM

- Collected and cleaned a corpus of Nepali poetry to fine-tune a multilingual base model.
- Added post-processing (rhyme and meter tweaks) to make generated verses feel more natural.
- Included evaluation notebooks tracking perplexity and stylistic diversity for guided tuning.

4. Air Quality Index(AQI) Prediction System

- Engineered features (lags, rolling averages) and trained regression models to predict AQI.
- Filtered noisy sensor readings to stabilize outputs and reduce spikes.
- Added simple explainability charts showing which pollutants drive changes.

5. Nepali SpellChecker

- Built a dictionary-based spell checking system for Nepali language using Python and Jupyter Notebooks for word verification and suggestions.
 - Compiled and cleaned a comprehensive corpus of Nepali words to enable accurate spelling validation.
 - Designed modular architecture for seamless integration with other Nepali NLP applications.
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6. EcoFarma

- Created an interactive farm simulation game (React/Python/FastAPI) with grid-based farming system for educational crop management.
- Implemented ML-based crop recommendation model analyzing soil parameters (N, P, K, pH, moisture) to suggest optimal crops with yield predictions.
- Built AI plant pathology assistant using Google Gemini for disease diagnosis from images with detailed treatment recommendations.

7. Nepal Connect

- Developed full-stack travel platform (React/Node.js/Express) with AI-powered trip planning using Google Gemini for personalized Nepal itineraries.
- Implemented intelligent chatbot for natural language planning with budget tracking, currency conversion, and day-by-day itinerary generation.
- Integrated Google Places API, geocoding, interactive maps, and Google Sign-in authentication for seamless booking experience.

8. KrishiBot

- Built AI-powered disease detection and soil type prediction system using TensorFlow with crop recommendations based on environmental parameters.
- Integrated 120-day weather forecasting with agricultural risk analysis, yield estimation, and harvest prediction using Open-Meteo API.
- Developed interactive dashboard (Next.js/TypeScript/FastAPI) with real-time monitoring, GDD tracking, and community Q&A forum for farmers.

9. JyotirVidhya - Udbodha Jyotish Website

- Developed a full-stack astrology service platform using PHP/CSS with responsive design for consultations and bookings.
- Implemented user-friendly interface with cross-browser compatibility and mobile responsiveness.
- Optimized page structure and styling for improved performance and user experience.

10. Portfolio Website

- Created a modular personal site where new projects can be added quickly.
- Improved performance using lazy loading and proper meta/OG tags.
- Integrated lightweight analytics to track which project sections get the most views.

PROFESSIONAL EXPERIENCE

Software Engineer, Berkeley Computer

Jan 2025-Present

- Built core web writing platform (JavaScript/PHP/CSS) with modular editor and reliable autosave, improving draft stability.
- Optimized frontend (code-splitting, asset + CSS consolidation) to speed initial page load and enhance user responsiveness.
- Introduced structured issue triage and lightweight CI (lint/build checks), reducing rework and accelerating feature delivery.

Co- Founder, Lekhnus Pro - AI Powered Nepali Writing Platform

June 2025-Present

- Co-founded AI writing platform for Nepali creators with custom Devanagari-optimized language models and real-time voice-to-text input.
- Built intelligent editor featuring inline AI suggestions, context-aware formatting, auto-save, and multi-format export optimized for Unicode Nepali.
- Launched freemium SaaS model serving journalists, marketers, and content creators across Nepal's digital ecosystem.

Data Engineer, Yagya Infotech

Aug 2025 - Sept 2025

- Designed unified JSON schema and validation scripts for large MCQ repository, preventing malformed or duplicate entries.
- Added versioned manifests and integrity checks (ID collision, duplicate stem detection) to improve dataset reliability.
- Automated ingestion and quality reports (coverage, difficulty distribution), cutting manual curation effort and speeding reuse.

ACHIEVEMENTS

Technical Member, Tech & IT Team,Kathmandu University Model United Nations 2025

- Built a responsive event site (HTML/CSS/JS) for schedules, committees, and registration info.
- Optimized images and assets to improve load speed and user experience.
- Set up automated deploy with basic link and HTML validation checks.

Event Manager, Interactive Googling Competition,ITMeet 2024, Kathmandu University

- Designed a KBC-style quiz system with a database-driven question repository.
- Enabled participants to access questions via mobile devices for real-time competition.

Editor BISMAYA, Annual Magazine and Monthly Newsletter of Kathmandu University Mathematics Students' Club (KUMSC)

- Built a responsive event site (HTML/CSS/JS) for schedules, committees, and registration info.
- Optimized images and assets to improve load speed and user experience.
- Set up automated deploy with basic link and HTML validation checks.

AgroTech Track Winner, MBM IdeaX 48-Hour Hackathon

- Led Team DevCoders to build EcoFarm, an AI-powered farming simulation game for sustainable agriculture education.
- Delivered complete full-stack solution (React 19/FastAPI/TensorFlow/Gemini AI) within 48 hours of intensive development.

EDUCATION

Bachelor of Data Science (BDSc)

Sept 2024 - Present

Kathmandu University, Dhulikhel ,Kavre

High School (+2)

2022-2024

Polar Star Secondary English Boarding School,Chitwan